LUKE THOMPSON

ASSOCIATE SOFTWARE DEVELOPER

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Portfolio: xylardark.github.io/Portfolio-Studio/u/admin/miraterra · GitHub: github.com/XylarDark

PROFESSIONAL SUMMARY

Software developer who works across front end, tooling, and hardware-adjacent code, and who uses AI coding agents every day with review and tests before anything counts. I write JavaScript/TypeScript and Python, use Git and GitHub Actions, and built evaluation-style checks — a headless runner that hard-fails on a false "done." At TruckSail I wrote C++ firmware for truck ECU devices (STM32, CAN bus). I have not worked on lab-instrument or soil-intelligence software, and I have not modernized instrument-control systems; I would expect to learn that stack on the team.

CORE SKILLS

Languages	JavaScript/TypeScript, Python, C++, C#, SQL, HTML/CSS
Testing & CI	pytest, Vitest, unit tests alongside code, Git/GitHub, GitHub Actions, pull-request review
Developer tools	Editor automation, done-gates, merge checks, build queueing, agent-tool indexes
Hardware & web	STM32 / CAN bus firmware, REST APIs, MySQL, SQL Server, Supabase (Postgres), React, Vite

SELECTED PROJECTS

DreamEngine — solo-dev automation harness Nov 2026 – present

Independent (solo) · agent orchestration, scheduled automation · private repository, available on request

- Built a done-gate (rebase, test rerun, merge-ready check), merge train, auto-resume, status notifier, and tool index (123 merged pull requests as of 12 Nov 2026). Parallel agent lanes run in local coding-agent sessions on a five-minute scheduled tick. A first end-to-end dream is in progress.

HomeWorld — AI-agent tooling and verification Feb 2026 – present

Independent (solo) · Unreal Engine 5.8, C++, Python · private repository, available on request

- Built an editor automation bridge: agents and scripts send a JSON `{command, args}` request and get structured JSON back (world info, asset checks, a per-row pass/fail probe); new command modules plug in.
- Built a headless test runner that hard-fails on five causes of fake passes (unreadable report, wrong engine, zero tests run, stale report, stale binaries), each with a regression test, and a build wrapper with one machine-wide queue so parallel agents can't collide. Source changes wait for my review.

Tether — phone app for reviewing computer work Nov 2026 – present

Independent (solo) · TypeScript, phone PWA · went live 11 Nov 2026 · private repository, available on request

- Phone PWA with push notifications and actionable decision cards that write answers back and can run declared actions. The phone is for reading, approving, and running declared actions — not for editing code or opening a public port.

Portfolio Studio and Shift Aug 2026 – present

React, TypeScript, Vite, Supabase / Phaser 3 · Portfolio Studio · Shift

- Portfolio Studio is a public React/TypeScript media-résumé site with a Supabase (Postgres) backend, per-employer pages, and GitHub Actions deploys. Shift is a shipped TypeScript browser/PWA game with 83 Vitest test files.

EXPERIENCE

Game Developer · OutFusion · Hamilton, ON Feb 2022 – Dec 2022

- Implemented game UI/UX from the design docs; built and tested C# REST API integrations for player data; integrated and tested game UI across mobile and desktop; partnered on bug verification and release quality.

Software Developer · TruckSail · Ancaster, ON Feb 2020 – Jan 2021

- Wrote C++ firmware for truck ECU devices (STM32, CAN bus) with defect finding for reliability; built JS/Bootstrap UIs and optimized SQL; maintained GitHub workflows and did cross-browser QA.

Software Developer & QA Tester · Evenica · Burlington, ON May 2018 – Sep 2019

- Built and tested client software (Andrew Peller, Young's Market Company, MM Realty International): T-SQL, C#, Linux, and Laravel (MySQL, PHP, JavaScript/jQuery, Vue.js).

Delivery Driver (Uber, retail delivery) · Hamilton, ON Jun 2022 – present

EDUCATION

Diploma, Computer Systems Technology — Software Development · Mohawk College · Hamilton, ON · 2019

Valid Ontario Class G licence · Open to hybrid work in Toronto, from Hamilton, ON